NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 9

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Which of the following is not a component of OpenStack?

a. Suse
b. Nova
c. Swift
d. All of these

Correct Answer: a. Suse

Detailed Solution: Nova and Swift are two of the many components of OpenStack (Please refer Lecture 41@3:18)

QUESTION 2:

Where is the sensed data sent for further processing in WSNs?

a. Client node
b. Central node
c. Collect node
d. Sink node

Correct Answer: d. Sink node

Detailed Solution: The sensed data is sent to sink node for further processing in WSNs (Please refer Lecture 42@4:04)

QUESTION 3:

Select the statement(s) that denote the advantages of cloud computing.

Statement I: Elasticity

Statement II: Pay-per-use

Statement III: Self Service

a. Statement I

b. Statement II

c. Statements I, II, and III

d. None of these

Correct Answer: c. Statements I, II, and III

Detailed Solution: The advantages of cloud computing are Elasticity, pay-per-use and self service (Please refer Lecture 42@5:58)

QUESTION 4:

Which of the following is a limitation of WSN?

a. Price

b. Post deployment maintenance

c. Battery lifetime

d. All of these

Correct Answer: d. All of these

Detailed Solution: Price, post deployment maintenance and battery lifetime are the main

limitations of WSN. (Please refer Lecture 42@11:01)

QUESTION 5:

The managerial role is played by _____ in sensor-cloud architecture.

a. End-users

b. Sensor-Cloud Service Provider

c. None of these

d. All of these

Correct Answer: b. Sensor-Cloud Service Provider

Detailed Solution: Sensor-Cloud Service Provider plays the managerial role in sensor-cloud architecture. (Please refer Lecture 42@14:29)

QUESTION 6:

The optimal composition of _____ is a management issue in sensor-cloud.

a. Logistics

b. Pricing

c. Caching

d. Virtual sensor nodes

Correct Answer: d. Virtual sensor nodes

Detailed Solution: The optimal composition of Virtual sensor nodes is a management issue in sensor-cloud. (Please refer Lecture 43@4:39)

QUESTION 7:

Who coined the term Fog computing?

a. IBM

b. CISCO

c. All of these

d. None of these

Correct Answer: b. CISCO

Detailed Solution: CISCO coined the term Fog computing. (Please refer Lecture 44@4:18)

QUESTION 8:

What are the issues related to the current cloud model?

a. Volume

b. Latency

c. All of these

d. None of these

Correct Answer: c. All of these

Detailed Solution: Volume, bandwidth and latency are some of the issues related to the current cloud model. (Please refer Lecture 44@10:30)

QUESTION 9:

Sensor data is processed in the cloud before it is sent to the fog.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Sensor data is processed in the fog before it is sent to the cloud. (Please refer Lecture 45@6:11)

QUESTION 10:

What is the type of received data according to which Fog nodes work?.

a. Non-time sensitive data
b. Very time-sensitive data
c. Less time-sensitive data
d. All of these

Correct Answer: d. All of these

Detailed Solution: Fog nodes work according to the type of data which are very time-sensitive, less time-sensitive and non-time sensitive data. (Please refer Lecture 45@11:39)

QUESTION 11:

Which of the following are the advantages of Fog?

a. Quick decision making

b. Nodes can be mobile

c. Reduces the risk of latency

d. All of these

Correct Answer: d. All of these

Detailed Solution: The advantages of Fog includes quick decision making, nodes

can be mobile and Fog reduces the risk of latency . (Please refer Lecture 45@20:57)

QUESTION 12:

Which of the following requires the analysis at the nearest node?

a. Non-time sensitive data

b. Most time-sensitive data

c. Less time-sensitive data

d. None of these

Correct Answer: b. Most time-sensitive data

Detailed Solution: Most time-sensitive data requires the analysis at the nearest node. (Please refer Lecture 45@15:46)

QUESTION 13:

Fog nodes cannot connect and leave the network when necessary.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Fog nodes can connect and leave the network when necessary. (Please refer Lecture 45@25:27)

QUESTION 14:

Which of the following is used by the fog applications to increase the speed of service accessibility?

a. Low power

b. Good network connection

c. Accelerators

d. Analytics

Correct Answer: c. Accelerators

Detailed Solution: Accelerators increase the speed of service accessibility. (Please refer to Page 260, Chapter 11, Introduction to IoT. S. Misra, A. Mukherjee, and A. Roy, 2020. Cambridge University Press.)

QUESTION 15:

Fill in the blank. The concept of _____ enables physical hardware to be shared among multiple entities.

a. Hardware virtualization

b. Software virtualization

c. Module virtualization

d. All of these

Correct Answer: a. Hardware virtualization

Detailed Solution: The concept of hardware virtualization enables physical hardware to be shared among multiple entities. (Please refer to Page 262, Chapter 11, Introduction to IoT. S. Misra, A. Mukherjee, and A. Roy, 2020. Cambridge University Press.)

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